Table 3.2: Chemical characteristic of stain or dye

Acid stain	Basic stain	Neutral stain
Chromogen of acidic stain is negatively charged, so it is also known as anionic stain.	Chromogen or coloured part of basic stain is positively charged, so it is also known as cationic stain.	It is a complex salt of dye acid with dye base.
Used to stain the positively charged component of microbial cell.	Used to stain negatively charged component of microbial cell.	It stains both positive and negative charged components of microbial cell.
Example: Eosin, Nigrosin, India ink, Acid fuchsin, Congo red.	Example: Methylene blue, Safranin, Malachite green, Basic fuchsin, Crystal violet	Example: Giemsa stain, Leishmanstain.

Infobits

On the basis of demonstrating the living or non-living status of microorganisms, some stains are classified as vital stains. These stains differentiate between living and non-living microbial cells. For example, Tryphan blue selectively colour dead tissues or cells.

Certain stains will give a different colour to the cell inclusion bodies from its original colour. Such stains are called metachromatic stains. Metachromatic granules of *Corynebacterium diphtheriae* contain polymerized inorganic polyphosphate responsible for metachromasia with Toluidine blue or Methylene blue.

Negative Staining

In negative staining, the background is coloured and bacteria remains colourless. It is because the acidic dyes are repelled by the negatively charged bacterial surface. The background gets stained and the cell remains colourless. This technique is useful for revealing the cell shape, size and demonstrating capsule (Figure 3.3).

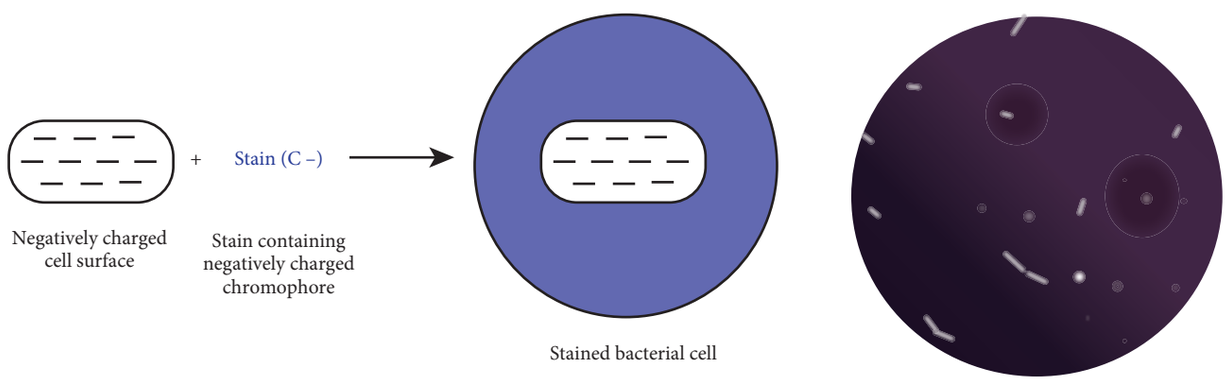


Figure 3.3: Negative staining

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